Table for Four — Project Scoping & Initial Agent Design

CMU AI Agent Certification — Capstone
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Document version: 1.2 · updated through the agent roster and the M4 governance layer
Status: Milestones 1–4 complete — search, booking backend, perks/RAG, LangGraph orchestrator, conversational concierge with long-term memory, web highlights, the declared **agent roster** with enforced grants (§6.3), and the **human gate + governance trail** (§6.4–6.5). 219 offline tests, no API key required. Remaining: the polished demo and the model-comparison stretch (M5).

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1. Executive summary

Table for Four is an agentic restaurant concierge. A user makes a natural-language dining request — *"Italian, near Midtown, 4 people, Friday 7pm, one guest is gluten-free"* — and the agent searches real restaurant data, reasons over which options fit, surfaces relevant perks (coupons/offers), and books a table through a reservation service. A **human-in-the-loop confirmation gate** sits before any booking is finalized, and a **governance/audit layer** records every tool call and approval.

The system is deliberately built **MCP-first** (tools exposed as Model Context Protocol servers) and **orchestrated with LangGraph**, so the reasoning loop, tool use, human approval, and auditing are explicit and inspectable rather than hidden inside a single prompt.

> ***Built with an agent, about an agent.*** *The project is itself scaffolded and developed using an agentic coding tool (Claude Code in VS Code) — a live, in-workflow demonstration of the same technology the capstone studies.*

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2. Problem statement

Booking a restaurant for a group is a small but real multi-constraint decision problem. A person must reconcile cuisine, location, party size, timing, budget, dietary restrictions, ratings, and availability — then act on it (book) — often across several tabs and apps. It is:

- **Multi-constraint:** several soft and hard constraints interact.
- **Retrieval-heavy:** the right answer depends on current, real-world data.
- **Transactional:** it ends in an irreversible action (a reservation) that a responsible system should not take without explicit human confirmation.

This makes it an ideal, well-bounded testbed for an agent that must **search, reason, personalize, and act under human oversight** — exercising the full agent loop rather than a single retrieval or generation step.

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3. Goals & non-goals

3.1 Goals

- Turn a free-text dining request into a **ranked, reasoned set of options** using **real** restaurant data.
- **Personalize** those options with relevant perks/offers via **semantic retrieval (RAG)** over an unstructured perks store.
- **Book** a table through a reservation interface, gated by **explicit human confirmation**.
- Produce a **complete audit trail** of tool calls, data provenance, and approvals.
- Keep the system **runnable offline with no API keys** for development, testing, and grading, while supporting **live data** when keys are provided.

3.2 Non-goals (explicitly out of scope)

- **Real transactional booking** against OpenTable/Resy/SevenRooms. These are partner-gated; we use a **self-built mock reservation backend** by design (see §7).
- Payment processing, real coupon redemption, or any real financial transaction.
- A production web/mobile UI. There is now a **Streamlit chat surface** (the demo and recording surface, carrying the reserve gate, the photo strip and the live governance panel) alongside a terminal REPL and a one-shot CLI — but it is a demonstration of the agent loop, not a product front end.
- Multi-city scale, real-time availability guarantees, or account/identity systems beyond a lightweight member-preferences notion (stretch goal).

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4. Target user & primary scenario

Primary user: an individual arranging a meal for a small group who wants a single natural-language request handled end-to-end.

Primary scenario (happy path): 1. User: *"Italian, near Midtown, 4 people, Friday 7pm, one guest is gluten-free."* 2. Agent searches restaurants, applies constraints (cuisine, area, price, rating, open-now), and returns a shortlist with reasoning. 3. Agent retrieves and attaches **relevant perks** ("2 of these 3 have offers that fit a gluten-free group of 4"). 4. User picks an option. 5. Agent proposes a booking and **pauses for explicit confirmation**. 6. On approval, the booking is finalized and a confirmation returned. 7. Every step — searches, perk matches, the approval, the booking — is logged.

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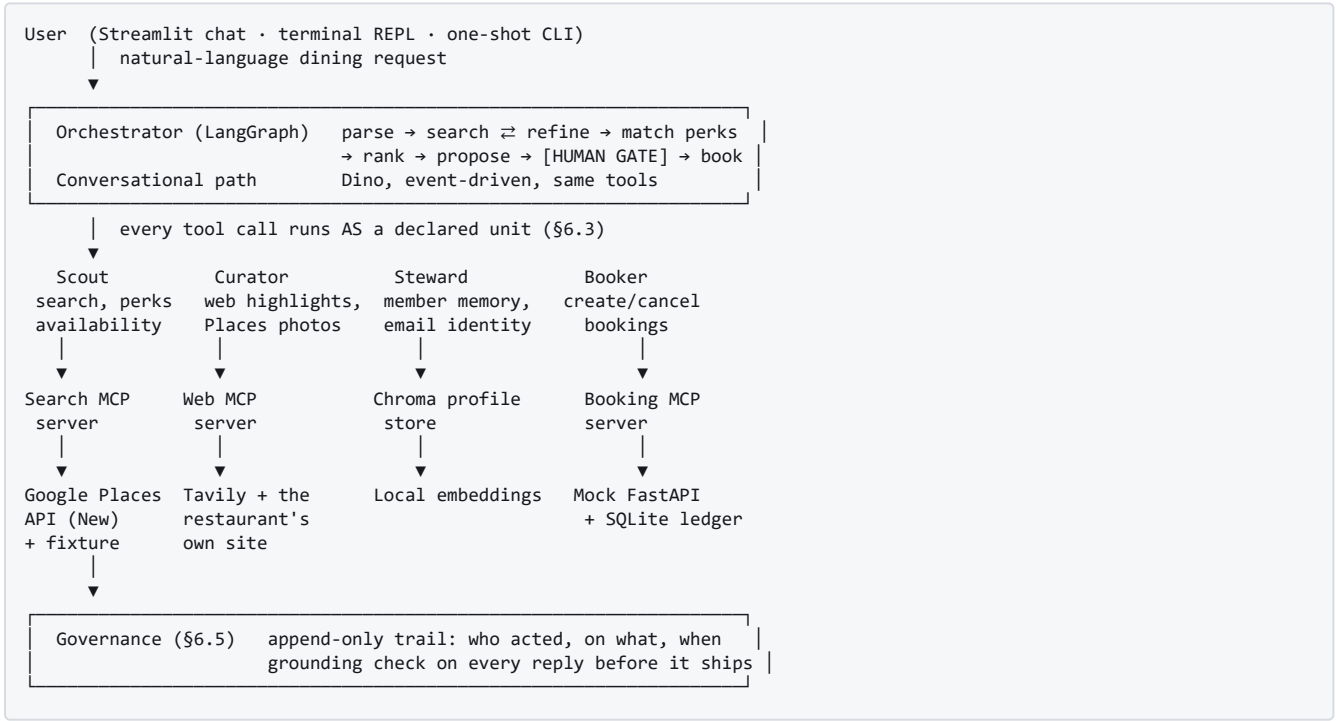
5. System architecture



Design principles - **MCP-first.** Every external capability (search, perks, booking) is a discrete MCP tool server. The orchestrator only knows tools, not vendors — swappable and independently testable. - **Offline-first, live-optional.** Each data path has a keyless offline mode (fixtures / synthetic data / mock backend) and a live mode behind an env var. The **same normalization code runs in both modes**. - **Provenance is first-class.** Every result carries a `source` field (`live` | `fixture` | `synthetic`) so the audit layer can record what a decision rested on. - **The irreversible step is gated.** No booking is finalized without an explicit human approval event — on **both** surfaces (§6.4), since the conversational one is where guests actually book. - **Authority is declared, not assumed.** Every capability belongs to a named unit (§6.3), and a unit reaching past its grant raises rather than proceeds. The component that holds the model holds no capability at all. - **A claim with an exact answer is**

checked, not estimated. Constraints are verified by function calls, and the outbound reply is checked against what tools actually returned (§6.5) — deterministically, so it costs no API key and cannot itself hallucinate.

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6. Agent design detail

6.1 Orchestration (LangGraph)

The agent is modeled as an explicit **state graph**, not a single prompt loop. Nodes as built:

Node	Responsibility
`parse`	Extract structured constraints from free text (cuisine, area, party size, time, dietary, budget).
`search`	Call the Search MCP tool; get candidate restaurants.
`refine`	On an empty search, relax **one** constraint and loop back to `search`. Bounded by `MAX_ITERATIONS` so it can never spin.
`match_perks`	Call the Perks MCP tool with the user's intent + candidate `place_id`s; attach fitting offers.
`rank`	Reason over fit; produce a ranked shortlist with rationale.
`propose`	Draft a booking for the chosen option.
`gate`	**Interrupts.** The run checkpoints and genuinely stops until a decision arrives from outside the graph. Only an explicit approval resumes it; a malformed resume, a missing approver, or an unattended run all decline. ✅ Built (M4)
`book`	On approval, call the Booking MCP tool; return confirmation.
`audit`	Close the run with a structured record — reached whether or not the booking went ahead, and stating plainly whether a human approved.

State is carried explicitly between nodes so the flow is inspectable and resumable — a core reason for choosing LangGraph over an implicit agent loop.

6.2 Tools (MCP servers)

A. Search — `search_restaurants` *(built, Milestone 1)* - Backed by **Google Places API (New)**, with an **offline fixture** fallback. - Normalizes raw results to a clean shape (`place_id`, `name`, `address`, `price_level` 0–4, `rating`, `open_now`, `location`, …). - Server-side-style filters: `max_price_level`, `min_rating`, `open_now`, `max_results`. - Uses an explicit **field mask** so live calls fetch only needed fields — this controls both the response shape and the **billing SKU** (cost control). - Returns `source: "live" | "fixture"`.

B. Perks — `find_perks` *(built, Milestone 2.5)* - Backed by a **Chroma vector database** of **synthetic** perks. - **Hybrid retrieval:** semantic vector similarity over an unstructured `blurb` (cuisine/vibe/dietary/occasion) **combined with** structured metadata filtering (`min_party_size`, `expiry`, `dine_in_only`, `perk_type`, `discount_pct`, `valid_days`, `active`). - **Local embeddings** (`all-MiniLM-L6-v2`) — no API key, fully offline. - Perks are keyed to restaurants by `place_id`. Returns match score and `source: "synthetic"`.

C. Booking — `create_booking` / `check_availability` / `cancel_booking` *(built, Milestone 2)* - Backed by a **self-built mock FastAPI reservation backend** (see §7). - Exposes availability lookup and reservation creation with a confirmation id. - Deterministic/mocked so the transactional path is fully testable offline. - Cancellation enforces the **24-hour policy in the backend**, not the model: inside the window it refuses and returns the restaurant's phone/website instead.

D. Web highlights — `lookup_dining_highlights` *(built, see §9.2)* - Backed by **Tavily** web search, with an offline fixture fallback (placeholder graphics) so the journey still runs with no key. - Returns cited menu highlights plus photos for a restaurant already recommended or booked — never an open-ended web search (that would breach the dining-only scope). - Scoped to the restaurant's **own domain first**, widening to the open web only when that returns nothing about the food. - Also **reads the restaurant's own site directly** for its published imagery (`og:image`, schema.org `image`), which a search index cannot be made to do — see §9.1 for why that ordering matters. - Feeds the **generated menu cards** (§9.1), whose dish lines are strictly extractive from retrieved text.

E. Restaurant photos — `place_photos` *(built, Milestone 3.6)* - Resolves the photo references Google Places returns against a `place_id`, so the images are of **that** restaurant by construction rather than by name match. - Fetched with `skipHttpRedirect` so the API key stays server-side: the media endpoint otherwise redirects straight to the image, which would publish the key in an `<img src>`. - Granted to Curator alone (§6.3), and narrow by design — it resolves handles Scout already produced for a restaurant already in the conversation, and cannot find a restaurant.

6.3 The agent roster — five declared units *(built, week 4)*

Not in the original scoping, and the most consequential structural addition since. The steps above differ in **authority**, not merely in skill: searching is free and reversible, writing to a member's profile changes how they are treated on every future visit, and creating a booking commits them to being somewhere at a time. A single component holding every capability has no way to express that difference — it can only be *told* to behave.

So capability is declared. Each unit is a markdown file in `agent/roster/` whose frontmatter states what it may do, what it is explicitly denied, and which model-facing tools it answers.

Unit	Role	Holds	Cannot
Dino	Host, front of house — the only unit with a model	*nothing*	every capability in the system
Scout	Reservations desk: finds the table, never takes it	search, perks, availability	book, cancel, write to memory
Curator	Kitchen research: knows the food, never meets the guest	web highlights, Places photos	search, book, write to memory
Steward	Keeper of the book: the member's own record	remember, adopt an email identity	search, book, reach the web
Booker	The pass: the only unit that commits the guest	create / cancel bookings	search, reach the web, adopt an identity

Two properties make this more than documentation. **Grants are enforced**: `_dispatch` runs each handler as its owning unit, and the tool registry raises `NotGranted` rather than proceeding, so misbehaviour is an exception rather than a policy violation. **Dino holds nothing**, which inverts the usual arrangement where the reasoning component owns every tool and is asked politely to behave.

The harness was built under a hard constraint: it must cost nothing. Unit briefs are read from files that were already being sent, and a test pins the tool-schema budget, so the measured cost is **zero additional model calls and zero additional tokens**.

6.4 Human-in-the-loop gate *(built, M4)*

The one irreversible action is a deliberate, auditable human decision. Because the system has two surfaces, it has two gates, and both refuse by default.

- **Orchestrator.** `gate_node` issues a LangGraph `interrupt`: the run checkpoints and stops until a decision arrives from outside. `run_concierge` has **no default that books** — no approver means declined, because a convenient default would quietly undo the gate.
- **Conversation.** Every validation runs, and then the booking **stops**: the full details go on screen (restaurant, address, date, time, party, name, email, the perk, any dietary note) with **Reserve** and **Change my mind**, and `create_booking` is not called until one is pressed. The summary is built from the same object the booking is built from, so what the guest approves cannot drift from what is written. A prose *"shall I confirm?"* is the model's reading of consent, not consent — this gap existed until it was closed by demo feedback.

A surface that cannot show a button (the terminal REPL) keeps the direct path: a gate nobody can answer is a dead end rather than a safeguard.

6.5 Governance & audit layer *(built, M4)*

`src/table_for_four/governance/` holds two things, both deliberately deterministic.

The trail (`audit.py`) is append-only, one record per consequential step, each carrying `event` , `actor` , `member_id` and a timestamp: which tool was called with what arguments, the data provenance (`live` / `fixture` / `synthetic`), the approval or decline, and the booking outcome. The roster made the valuable part cheap — `_dispatch` already knew which unit was acting, so **every tool call names who ran it**. Records are held per session and optionally mirrored to JSONL; the Streamlit sidebar shows the trail filling in live, so the claim is demonstrated rather than asserted.

The grounding check (`grounding.py`) was not in the original scoping and closes a real gap. The tool boundary refuses to *act* on an invented detail, but nothing stopped the model *saying* one. Every reply is now checked before the guest sees it: each **time, date, confirmation id and email** must trace to an actual tool result, or the sentence carrying it is removed and the removal recorded. It is code rather than a second model, because these four have exact answers already in session state — which also means it costs no API key and cannot itself hallucinate. What it deliberately does **not** cover is named plainly rather than implied away: dish and restaurant names need entity extraction and a judgement, and a street address reads too much like a phone number to risk deleting the sentence around it.

6.6 Memory model

The agent uses three distinct tiers of memory; naming them separately keeps their lifetimes and responsibilities clear.

Tier	Holds	Implementation	Milestone
Working memory	The in-flight request: parsed constraints, candidate restaurants, matched perks, the chosen option, the pending booking.	LangGraph **typed** `State` carried across nodes, persisted by a **checkpointer** keyed per conversation **thread**. Enables the human-gate **pause/resume**.	M3 / M4
Long-term profile memory	Member preferences: name/pronouns, email, dietary defaults, favored cuisines, party size, kids, and past bookings.	A **Chroma** store keyed by email, with two retrieval modes (key lookup + semantic recall). Read when a guest is recognised, written as they talk. Cuisines are a **rolling window of the last three** — taste drifts — while dietary needs are never aged out. Three fields describe the *member* rather than the outing — **home area, usual party size, favourite cuisines** — and are **consent-gated**: a first value is learned freely, but once on file it only changes when the guest agrees, offered once — either alongside the booking confirmation or the moment a returning guest is recognised by email, whichever comes first. Silence changes nothing.	✅ Built (M3.5)
Audit / episodic memory	Every consequential event: tool call + args, data `source` , human approval/decline, booking outcome.	Append-only **governance trail** (§6.5), naming the acting unit on every call; doubles as booking history.	✅ Built (M4)

6.7 Control flow & reasoning loops

The orchestrator is a **state graph**, not a single prompt — control flow is explicit and inspectable.

- **Happy path (linear):** `parse → search → match_perks → rank → propose → [gate] → book → audit` .
- **Reasoning loops (conditional edges):** the graph can **refine and retry** — e.g. if `search` / `match_perks` return nothing that satisfies the constraints, `rank_and_explain` may relax or rewrite the query and route **back** to `search` . A **max-iteration guard** bounds this so the loop cannot spin. This refine-retry cycle is where the agent's reasoning is exercised, beyond a single linear pass.
- **Human interrupt:** at `gate` the graph **interrupts** and genuinely stops; on approval it **resumes** from the checkpoint, on decline it routes to the end without writing. The pause/resume is powered by the working-memory checkpointer. The conversational path has its own gate (§6.4), enforced in the handler rather than the graph, because that path is event-driven and has no fixed sequence.

- **Tools as a registry:** each capability is an independent **MCP server** and the orchestrator is an **MCP client**. Adding a capability means adding a server, not rewriting the graph — and, since week 4, granting it to a unit (§6.3). Current tools: `search_restaurants` (M1), `find_perks` (M2.5), `check_availability` / `create_booking` / `cancel_booking` (M2), `lookup_dining_highlights` (M3.6) and `place_photos` (M3.6).

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7. Data strategy

Capability	Source	Rationale
Restaurant search	**Real** — Google Places API (New), with offline fixture	Real-world retrieval is core to the value; fixture keeps it keyless/testable.
Perks / coupons	**Synthetic**, clearly labeled	No cleanly-licensable free coupon dataset exists; scraping real coupon sites is ToS-gated and weakens the governance story. Labeled synthetic data is the more defensible, reproducible choice.
Booking backend	**Self-built mock** (FastAPI)	Real reservation systems (OpenTable/Resy/SevenRooms) are partner-gated; Yelp dropped its free tier. Mocking a partner-gated downstream is a standard, defensible agent-prototyping pattern.

Provenance honesty. Every payload carries an explicit `source`. The system never presents synthetic or mock data as if it were live — this labeling *is* part of the responsible-AI design, not an afterthought.

Synthetic perks generation. A seeded (reproducible) generator produces several perks per fixture restaurant, spanning perk types, with some deliberately **expired** and some **party-size-gated** (to prove the metadata filters work) and varied dietary/occasion themes (to give semantic retrieval real signal).

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8. Technology stack

Concern	Choice	Why
Language / runtime	Python ≥ 3.12, managed by **uv**	Fast, reproducible, lockfile-based env.
Tool protocol	**MCP** (`mcp[cli]` , FastMCP)	Standard, inspectable tool interface; decouples agent from vendors.
Orchestration	**LangGraph**	Explicit, resumable state graph with a native interrupt for the human gate.
Vector store	**Chroma** (local, persistent)	Simple local RAG with built-in embeddings; no external service.
Embeddings	`all-MiniLM-L6-v2` (local)	Free, offline, no API key — consistent with offline-first philosophy.
Booking backend	**FastAPI** (mock)	Lightweight, self-contained transactional service.
HTTP	`httpx`	Async-capable modern client for the Places call.
Config	`python-dotenv`	Keeps keys out of code and version control.
Testing	`pytest`	Offline test suites per component.

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9. Milestone plan

#	Milestone	State
1	Search MCP server (Google Places, offline-testable)	✅ Complete
2	Mock booking FastAPI + booking MCP server	✅ Complete
2.5	**Perks / RAG:** synthetic perks → Chroma → `find_perks` MCP tool	✅ Complete
3	LangGraph orchestrator; end-to-end happy path (search → perks → book)	✅ Complete
3.5	Conversational concierge (Dino) + long-term member memory in Chroma	✅ Complete
3.6	Web highlights via Tavily + generated menu cards (§9.1, §9.2)	✅ Complete
3.7	**Agent roster** — five declared units, enforced grants (§6.3)	✅ Complete
4	**Human gate on both surfaces + governance trail + reply grounding** (§6.4–6.5)	✅ Complete
5	*(Stretch)* model comparison, polished demo	

Two things arrived that the original plan did not name. The **roster** (3.7) came out of week 4 and reorganised how every other milestone's capabilities are held. The **grounding check** shipped inside M4 because building the trail made the gap obvious: the tool boundary refused to act on invented details while nothing checked what was said.

9.1 Restaurant imagery *(built, delivered differently than scoped)*

The intent — a richer visual surface that respects representational integrity (§10) — shipped, but not as a photo library. Two layers, both avoiding the original plan's weakness (a folder of stock images that must never be mistaken for a real venue): - **Generated menu cards** (`src/table_for_four/agent/menu_card.py`): one cuisine-themed SVG card per restaurant, carrying the retrieved dish highlights and the perk. **Six themes**, rendered as data URIs — no image files, no storage question, works offline. Nothing depicts a specific venue, so nothing can misrepresent one. - **Real photos, when they exist**, in a strict order of how firmly each can be tied to *this* restaurant: **Google Places photos** first (keyed to the `place_id` , so right by construction), then images published on the **restaurant's own domain** (read from its `og:image` and schema.org markup), then photos lifted from pages that matched a domain-scoped search, and only last a general image search. Each is captioned with where it came from. The ordering is the whole point: a search index has only the *name* to go on, which is how a guest ends up looking at another branch, or a namesake in another city. - **Honesty rule:** every dish line on a card is a **substring of retrieved text**. Where a menu isn't published online, the card says so rather than inventing a plausible dish. - **Storage:** unchanged from the original scoping — image bytes are **never** put in the vector DB. - *(Optional stretch-within-stretch:* multimodal CLIP embeddings to enable visually-similar retrieval — the one design that would let images genuinely earn a place in the vector store.)

9.2 Web enrichment via Tavily *(built, ahead of M4)*

Shipped as `lookup_dining_highlights` (§6.2 D) — the tool that adds the *qualitative* context Google Places doesn't carry: signature dishes, what a room looks like, what diners keep mentioning. Built earlier than planned because it is what makes the chat feel like a concierge rather than a booking form. - **Scope:** enrichment only; it does **not** replace Places for core search (Places is more structured for constraint matching). - **Bounded to the conversation:** it will only look up a restaurant already recommended or booked in this session, so it cannot become a general web-search back door around the dining-only guardrail. - **Offline behaviour:** rather than being live-only as first scoped, a fixture path returns seeded highlights with locally generated placeholder graphics, so the whole journey still demos with no key. - **Representational integrity:** every snippet carries its source domain, photos are captioned with where they came from and never presented as the restaurant's own, and the model is instructed to attribute dishes ("diners keep mentioning…") rather than promise them, since menus change.

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10. Responsible AI & governance considerations

- **Human authority over irreversible actions.** Bookings require explicit confirmation; the agent proposes, the human disposes.

- **Data provenance & honesty.** `source` labeling on every payload; synthetic and mock data are never disguised as live.
- **Auditability.** A structured trail of tool calls, sources, and approvals makes every decision explainable after the fact.
- **Cost control & least-privilege data access.** The Places field mask requests only the fields the agent reasons over, minimizing both data collected and billing.
- **Secret hygiene.** API keys live only in a gitignored `.env`; the recommended key is restricted to the single API it needs, capping blast radius if leaked.
- **Reproducibility.** Offline modes and a seeded synthetic generator make runs deterministic and gradable without credentials.
- **Representational integrity (imagery).** *(Built, §9.1.)* Generated menu cards depict no venue at all and are labeled illustrative; real photographs are shown only where they can be tied to the restaurant by construction — **Google Places photos**, keyed to the `place_id`, and images published on the **restaurant's own domain**, with an open-web image search used only as a last resort. A redirect that leaves the requested domain is refused outright, because a lapsed restaurant domain can be resold and its new owner's artwork would otherwise be shown as the restaurant's own. Captions name the actual source rather than claiming a single generic provenance.
- **Curated, not quoted.** What is written to a member's permanent file is normalized before it lands: an area is stored as a place ("Soho") rather than the sentence it was asked in, and a phrase naming nowhere ("nearby") is refused rather than filed. The normalizer only ever deletes words, so it cannot invent a neighbourhood the guest never mentioned — a guarantee a fluent rewrite could not make.

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11. Risks & mitigations

Risk	Mitigation
Live vs. synthetic **join-key mismatch** (Places `place_id`s differ from fixture ids, so perks won't match live results)	**Resolved.** Offline, perks key to the fixture set; live, a cuisine-matched **sample offer** is attached to the top results and labeled illustrative, so the perk story survives real Google ids without pretending a real offer exists.
Vector DB used for its own sake (perks that are purely structured don't need embeddings)	Hybrid design: embed genuinely unstructured blurbs, filter on structured metadata — vector search only where semantics add value.
Heavier dependency (`chromadb` pulls in onnxruntime)	Accepted; local embeddings avoid an external service and keep the offline story intact.
Over-automation of the **irreversible booking** step	Non-negotiable human gate before any write; declines are first-class.
Misrepresentation via imagery (a fabricated image implying it depicts a specific real restaurant)	Illustrative-only, clearly-labeled generic imagery offline; real Google Places photos only in live mode; never caption a synthetic image as a specific named venue (see §10).
Scope creep from stretch ideas	Milestones are ordered; stretch items (M5) only after the core loop works end-to-end.
An instruction standing in for a control — the model told not to do something, and doing it anyway	Repeatedly observed in testing: a date decided for the guest, a chosen time re-offered, consent read into prose. Each was fixed by changing what the model *has* rather than what it is told: withhold the list, refuse in the handler, put a button in front of the write. Where an instruction and a data shape disagree, the data shape wins.

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12. Success criteria

The capstone is successful when: 1. ✅ A natural-language request flows end-to-end: **parse → search → perk match → proposal → human approval → booking confirmation.** 2. ✅ The system runs **fully offline**

with no API keys (fixtures + synthetic perks + mock backend) and **switches to live search** when a key is present. 3. ✅ **Perk matching demonstrably uses both** semantic similarity and metadata filters (e.g. a gluten-free group-of-4 query surfaces a fitting, non-expired, party-size-valid offer). 4. ✅ No booking is ever finalized without a recorded **human approval** — enforced on both surfaces, with no default that books. 5. ✅ A **governance/audit trail** exists for every consequential step with its data provenance, and names the **unit that acted**. 6. ✅ Each component ships with **passing offline tests** — 219 of them, no key needed.

Two criteria the original list did not anticipate, added because building the system made them matter:

1. ✅ **A unit can only do what it was granted.** Capability is declared per unit and enforced at the tool registry, so a boundary is a raised exception rather than a remembered rule.
2. ✅ **The reply is checked, not just the action.** Times, dates, confirmation ids and emails in what Dino says must trace to a tool result, or they do not reach the guest. The coverage gap (dish and restaurant names) is stated rather than implied away.

End of document — v1.2.